

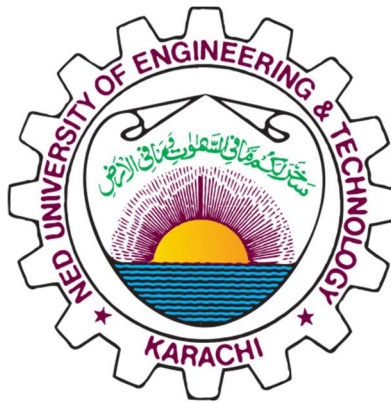
LAB REPORT

Digital Signal Processing

(EE-394)

Department of Electrical Engineering

NED University of Engineering and Technology



Open ended lab 1

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Objective:

To convert an analog (voltage & current) signal into digital signal using ADC (audio card) and display it on MATLAB Simulink environment.

Components:

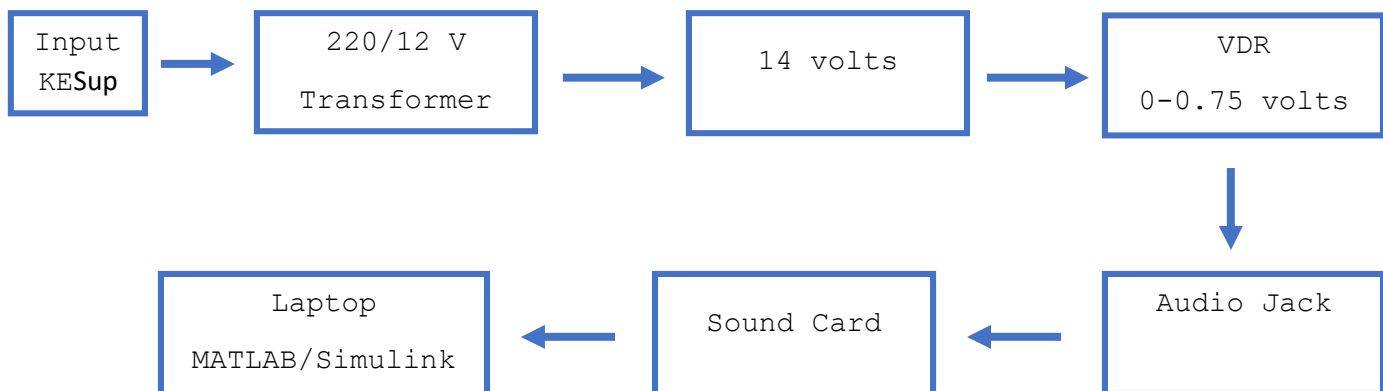
1. Audio card
2. Transformer (220V/12V)
3. Resistors (for VDR)
4. Verobaord
5. Audio Jack
6. PC with MATLAB environment
7. 100 Watt Bulb

Procedure:

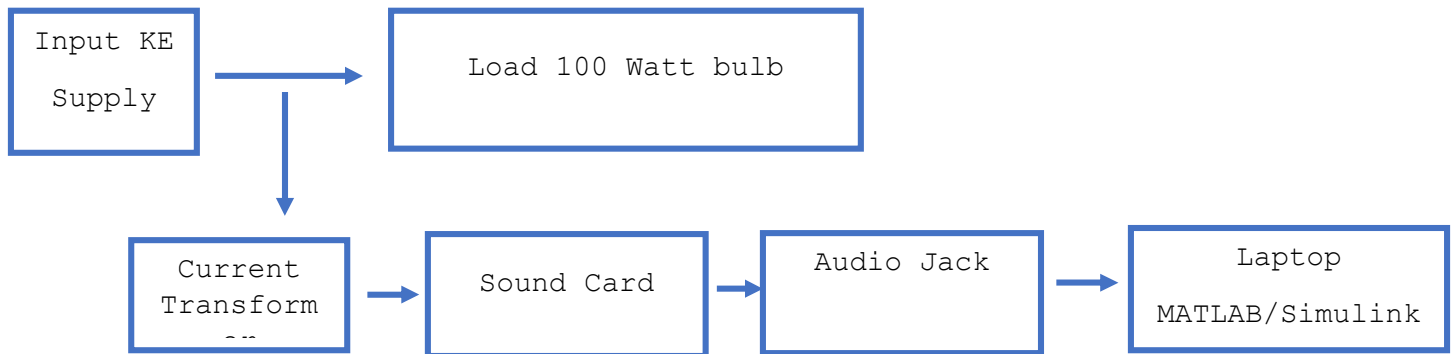
- Using transformer convert 220VAC from mains to 12VAC
- Using VDR convert 12VAC to a voltage compatible to audio
- Set the sampling frequency of the audio card ADC in MATLAB Simulink environment with proper justification
- Plot the acquired voltage waveform to Simulink scope
- Mention the safe operating range of your equipment

Block Diagram:

For Voltage:



For Current:



Project Summary:

In this project, we use VDR (Voltage Divider circuit) to reach the optimum permissible range of our sound card (ADC). In this case, we have a sound card whose voltage range should not exceed about 1 volt. We design VDR in such a way that the maximum voltage is about 0.8V. We set the least voltage possible so that our sound card doesn't get affected. We connect the components in the designed configuration. We plot the waveform on the Simulink time scope by using the respective block.

Then we adjust the obtained voltage wave to the actual 220 (KE) by using the gain block in the Simulink block diagram. We set the gain value at 2400. And hence, we obtain our actual waveform in Simulink. We then confirmed the calculated voltage parameters with the plot.

In order to obtain the current waveform, we use a 100 W incandescent bulb. For measuring the current, we use the CT (current transformer) clamp model and insert its output into the sound card's input. Then we reset the gain value to 2400 in the Simulink block to plot the current waveform of a 100W load. Since the load was entirely based on resistive load characteristics, the result we obtained was found to be:

$$I_{rms} = 100/240 \text{ (Wattage/Voltage rms)} \rightarrow 0.416 \text{ Amps.}$$

Then we compare it with our plot and found that the result was almost same. The other main constrain we need to do is to confirm that the frequency of both voltage and current plot is equal to 50 Hz, or in other words the time period for each wave is equal to 20ms.

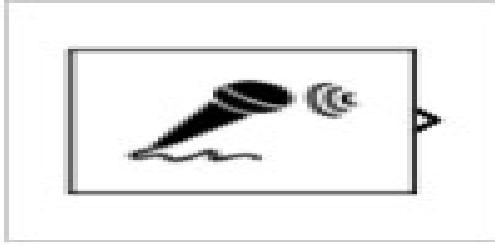


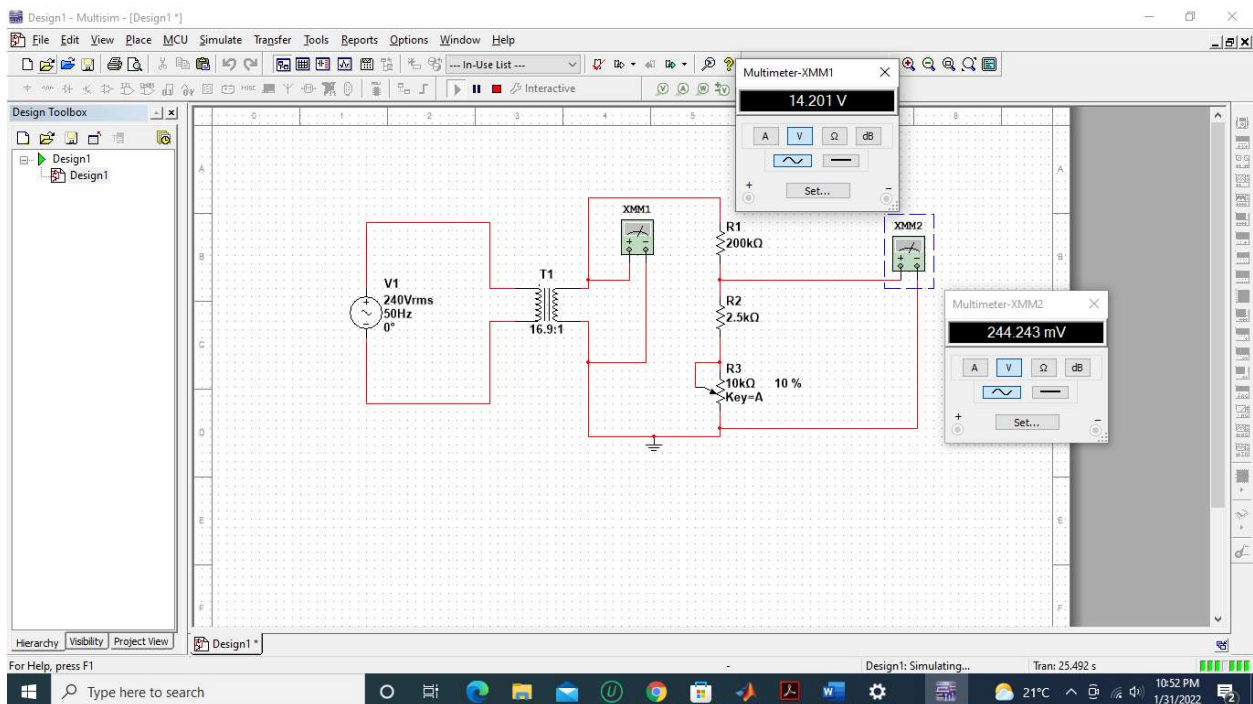
Figure 1: Audio device

Figure 2: time

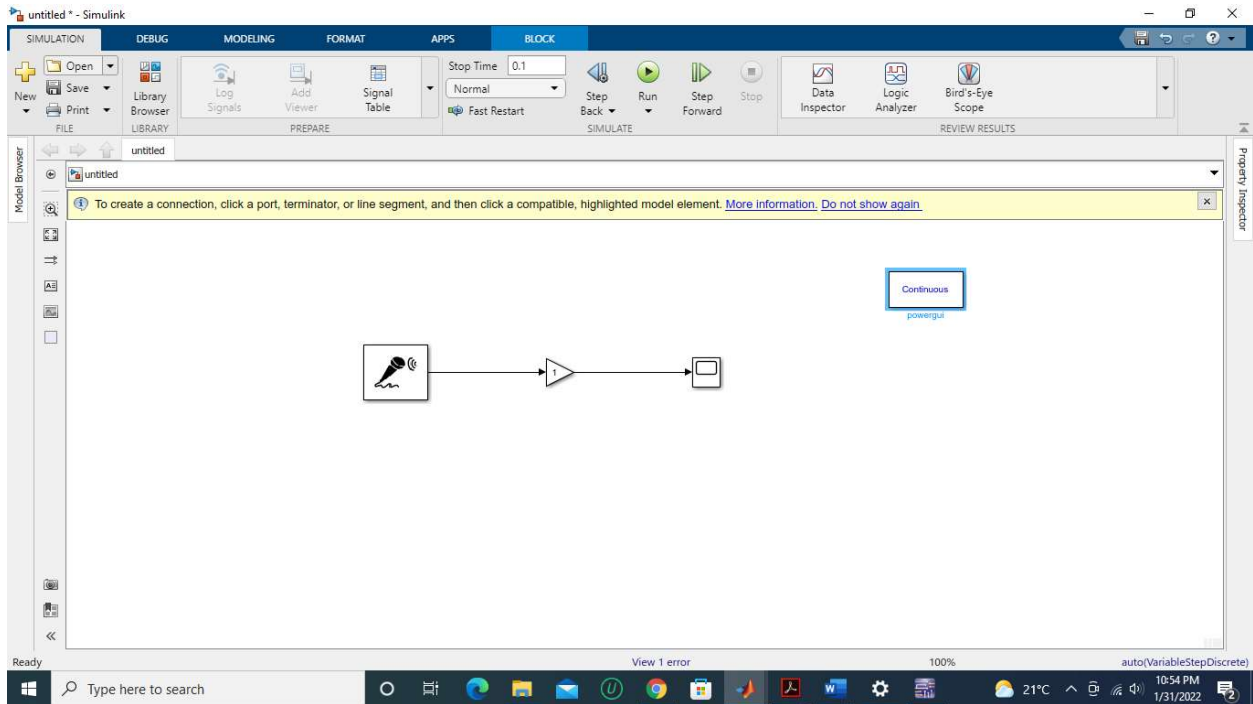
Complications:

- Sound card was having some issues initially then after discussing with classmates we change the sound card and get proper Sine wave, previously there is some distortion in Sine wave.
- Audio jack issues
- First, we connect audio jack to VDR circuit using alligator clips but couldn't get proper results. Then soldered soft wires but it still there are some complications, in the end we soldered hard wires to audio jack and connect it to VDR circuit. Hence got better results.

Schematic of VDR:



Simulink Sketch:



Calculations:

VDR:

$V_s = 14.2$ Volts

$R_1 = 200K$ OHMS

$R_2 = 2.5K$ OHMS

Variable Resistor(POT) = $10K$ OHMS

$V_{out} = V_s \cdot (R_2 + VR) / (R_1 + R_2 + VR)$

$V_{out} = 0.17 - 0.83$ volts

Gain Factor:

For voltage:

Gain = input/output

Gain = $240 / 0.1 = 2400$

Current:

Input current = 0.416 rms

Resolution of ADC:

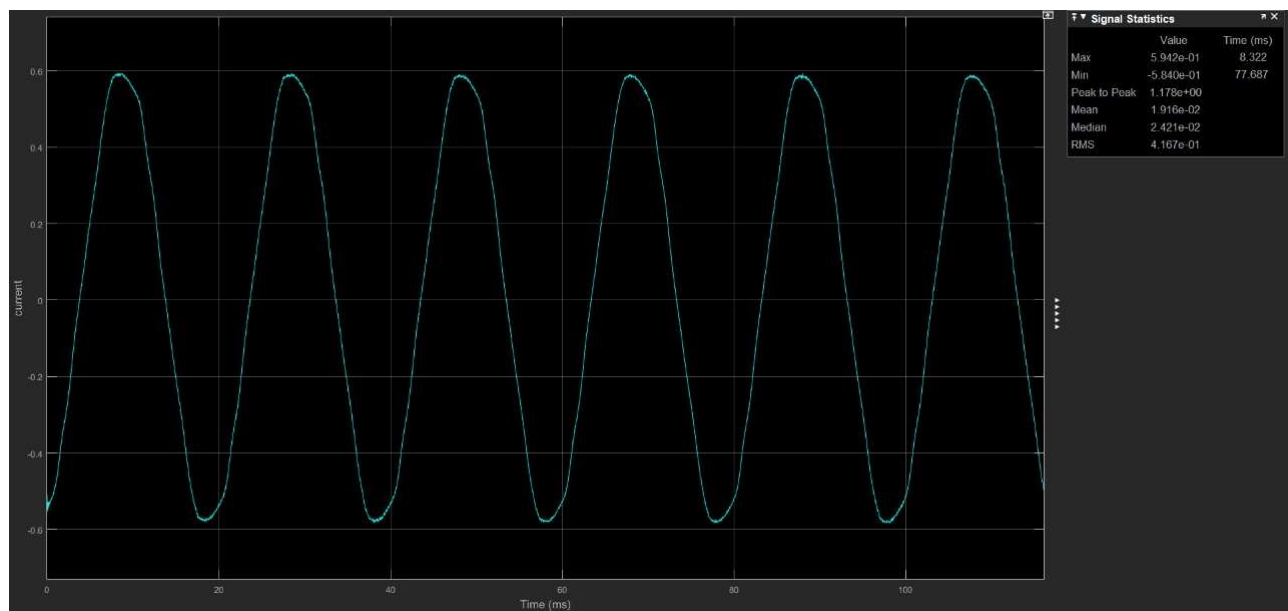
The resolution of ADC was found to be $15\mu\text{V}$

Sampling Frequency:

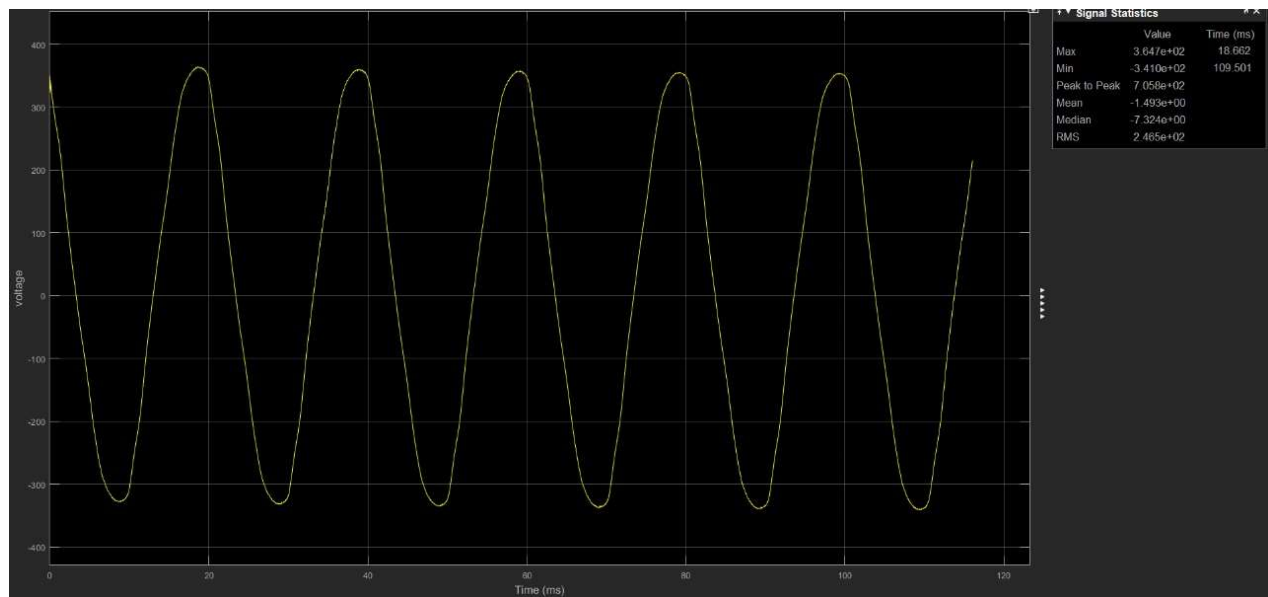
48000Hz

Plots:

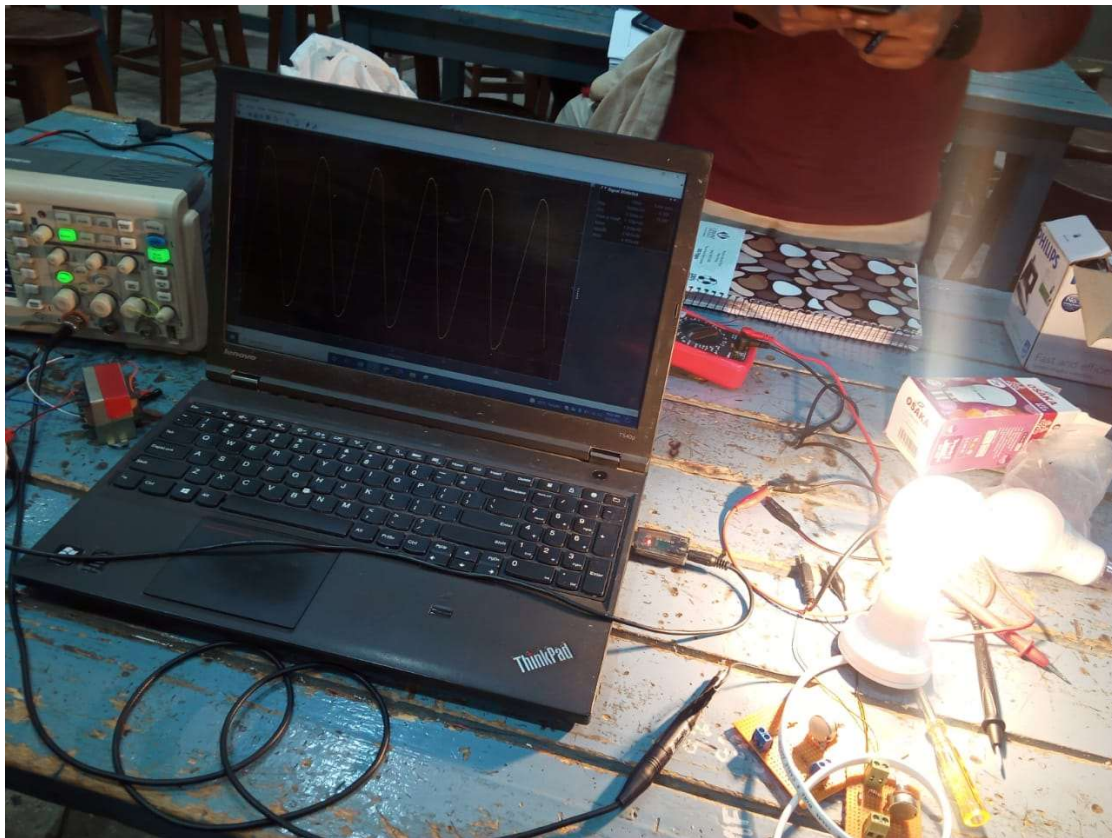
Current Plot:

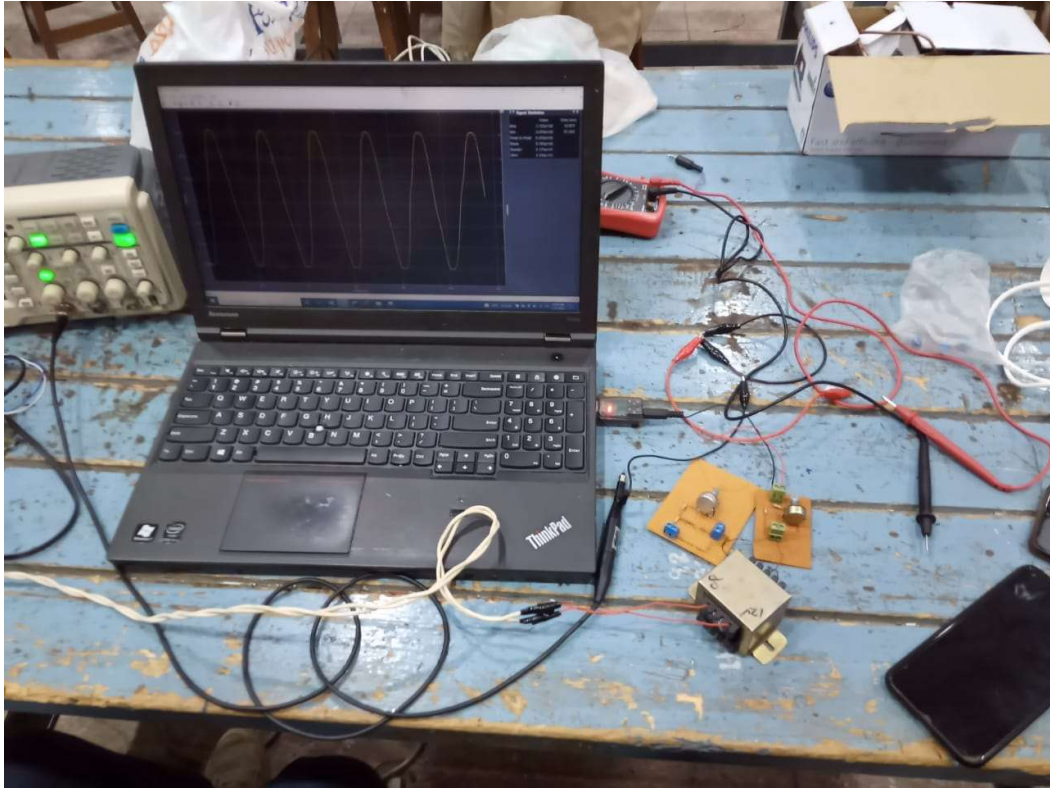


Voltage Plot:



Real Time Images:





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OPEN ENDED LAB 2

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Objective:

To convert analog (Voltage and current) signal into digital signal using ADC (audio card). Display it on MATLAB Simulink environment and perform Spectral Analysis of the resulting current signal.

Required Components:

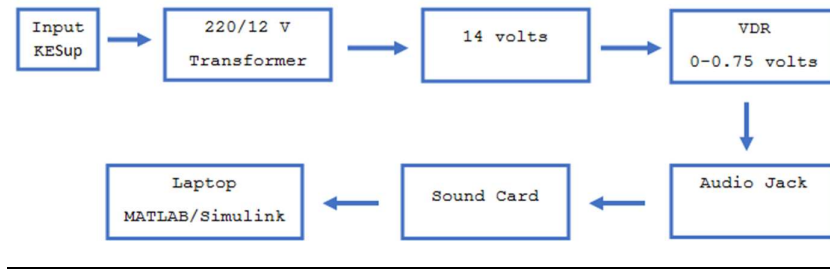
1. Audio Card
2. Current Sensor (current sensing resistor / hall effect sensor / CT)
3. Vero board
4. Audio jack
5. Harmonic producing Load (Electronic devices)
6. PC with MATLAB environment
7. 12 V Led Bulb.

Procedure:

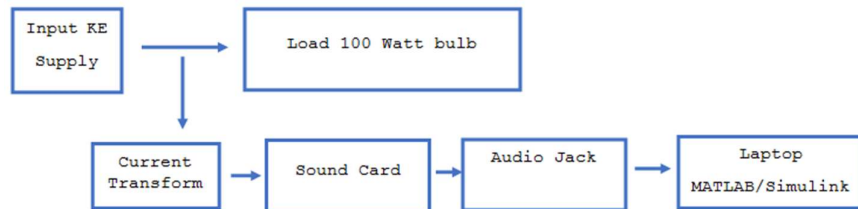
- Using current sensor, convert the current flowing through load into an equivalent voltage.
- If required, using VDR to convert the voltage as obtained from current sensor to a voltage compatible to audio card (show all the calculations of resistances with their power ratings).
- Set the sampling frequency of the audio card ADC in MATLAB Simulink environment with proper justification
- Plot the acquired current waveform to Simulink scope.
- Mention the safe operating range of your equipment.
- Plot the frequency spectrum of the obtained current and Voltage waveform. Use windowing function to reduce DFT leakage if required.
- Also, plot the frequency spectrum of the line voltage as obtained in open ended lab 01.

Block Diagram:

For Voltage



For current



Project Summary:

In this project, we purposefully used the CT (current transformer) to plot the waveform of the current wave and OSAKA 12.1 W Led Bulb to create harmonics in the current wave. The reason behind the creation of harmonics is due to the ballast circuit installed in the hood of led bulb. While, the rest of the circuit is same as in lab 1 and carried forward.

We have used the following blocks in Simulink:

- 1) Audio device reader
- 2) Time Scope

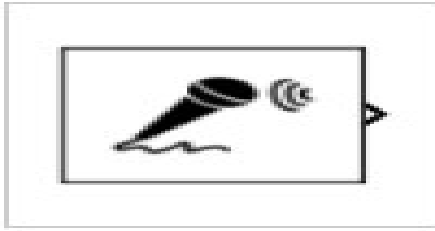


Figure 1 Audio Device Reader

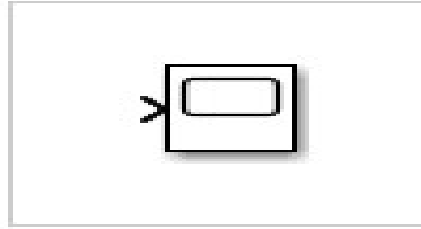


Figure 2. Time scope

Then we obtained the harmonic containing distorted sinusoid in the current waveform.

The main objective behind this project is to apply the Fast Fourier Transform on the distorted waveform to get the correct fundamental frequency whose harmonics are produced in the wave. We devised the MATLAB code to do such task in real time. The FFT application results in the plot showing the variation of plot at 50 Hz frequency. This little disturbance in FFT is due to the fact that integral numbers of harmonics of 50 Hz are also present in the plot.

In order to avoid the undesirable frequencies, we use MATLAB's Windowing command to create smooth multiple windows (Blackman, Kaiser and etc) and created the desired output of FFT.

One thing about the magnitude of FFT (Y-axis) in MATLAB's algorithm; it doesn't only depend on the strength of the input signal but also the number of samples we are using in the real time plot. We adjusted the whole formula in such a way that the plot of FFT gets correct magnitude and fine frequency.

Complications:

1. Identification of suitable harmonic load and its selection.
2. Understanding the algorithm of FFT in order to develop MATLAB code for our project.
3. We faced the sound card issue again and trouble-shoot it to sort it out.
4. Adjusting the magnitude of current transformer in order to get exact current.
5. We were stuck in an ambiguous situation. We calculated the power factor of our LED Lamp in electrical machines lab.

MATLAB Code:

```
clear;

clc;

close all;

Fs=48000; bits=16 ;channel=1;

t=1/Fs;sec=0.1;

Ts=t:t:sec;

while 1

zeeshan = audiorecorder(Fs,bits,channel);

recordblocking(zeeshan,sec);

y1=getaudiodata(zeeshan);

gain = (10*y1);

y = gain;


subplot(321);

plot(Ts,y1,'r');

xlabel('time'),ylabel('Amplitude'),title('Input Signal');


subplot(322);

plot(Ts,y,'r');

xlabel('time'),ylabel('Amplitude'),title('Input Signal with gain');


subplot(323)
```

```

f=fft(y);

N = length(y);

n = 0:N-1;

df = Fs/N; %frequency resolution

frequency_vector = n*df;%freq vector

ff =2*abs(f)/N; %fft= fft(+)=fft(-) therefore divided by 2 to get result

stem(frequency_vector,ff,'m','filled'),xlim([0 Fs/480]);

xlabel('frequency'),ylabel('magnitude'),title('FFT of Input Signal');


subplot(324);

w = window(@triang,N);

xw = y.*w; % windowing of original signal

plot(n,w,'k');

xlabel('signal length'),ylabel('magnitude'),title('Window');


% subplot(324)

% stem(abs(fft(w)),'c','filled'),xlim([0 100])

% xlabel('Samples')

% ylabel('Magnitude')

% title('Spectrum of Window Function')

% grid

subplot(325);

Xw = 2*(abs(fft(xw)))/N;

```



```

Lxw = length(Xw);

stem(frequency_vector,Xw,'g','filled'),xlim([0 Fs/480]);

xlabel('frequency'),ylabel('magnitude'),title('FFT of Windowed Signal with freq');

subplot(326);

plot(Ts,xw,'b');

xlabel('time'),ylabel('magnitude'),title('Windowed Signal with time');

end

```

Calculations:

For Current:

$$P = 12 \text{ W} \quad V = 240 \text{ V} \quad \text{pf} = 0.5$$

$$I = P/V * \text{pf}$$

$$I = 12 / (240 * 0.5) = 0.1 \text{ A}$$

VDR:

$$V_s = 14.2 \text{ Volts}$$

$$R_1 = 200\text{K OHMS}$$

$$R_2 = 2.5\text{K OHMS}$$

$$\text{Variable Resistor(POT)} = 10\text{K OHMS}$$

$$V_{\text{out}} = V_s * (R_2 + V_R) / (R_1 + R_2 + V_R)$$

$$V_{\text{out}} = 0.17 - 0.83 \text{ volts}$$

Gain Factor:

For Voltage:

$$\text{Gain} = \text{input/output}$$

$$\text{Gain} = 240/0.1 = 2400$$

For Current:

Input Current = 0.1.

Output Current (read by MATLAB) =0.01.

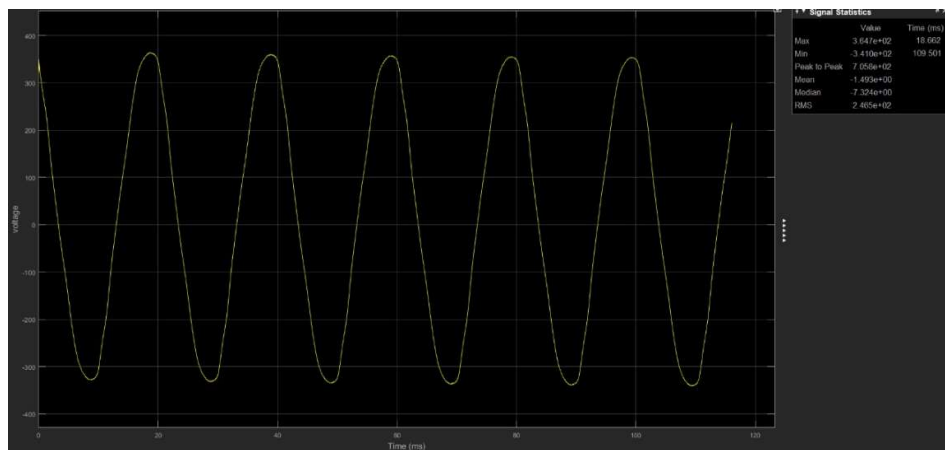
Gain = input/output.

Gain = 0.1/0.01.

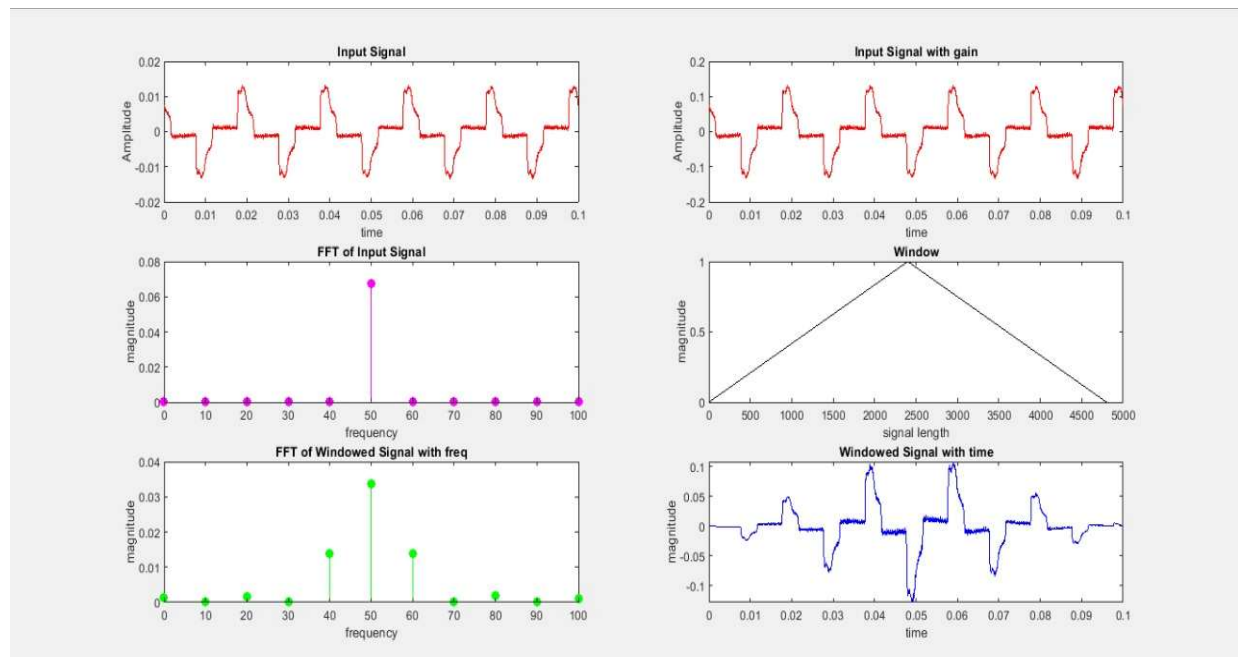
Gain = 10.

Plots:

For Voltage



For Current



Real project Image:

